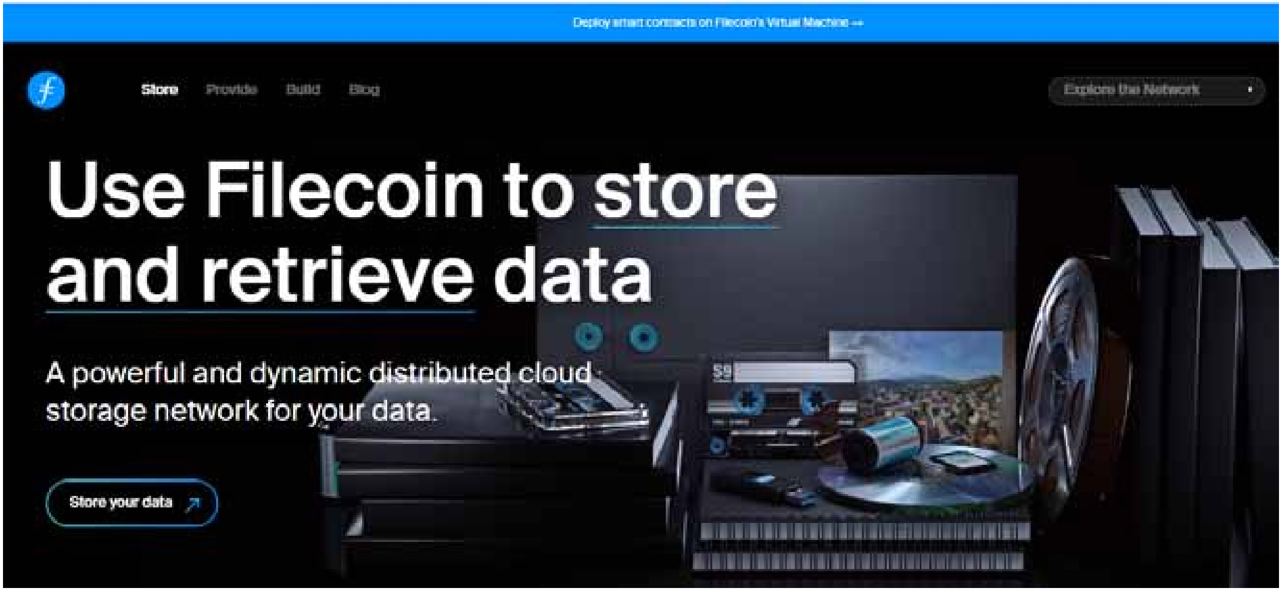

Figure 3: Filecoin platform in blockchain

One of the core pillars of blockchain technology is its immutability – the quality of being unchanging over time. Blockchain maintains the integrity and security of stored data through its immutable nature. Each block within the chain contains a cryptographic hash, a unique digital signature derived from the data in the block, including a reference to the previous block's hash. This chaining mechanism creates a chronological sequence, ensuring that any attempt to alter or tamper with data within a block would necessitate altering subsequent blocks in the chain, thereby alerting the entire network to the unauthorised change. This feature makes the blockchain extremely resilient to unauthorised modifications, making data tampering exceedingly difficult.

The immutability of the blockchain ensures that once data is stored and confirmed within the chain, it remains secure and unaltered. Any modifications or tampering attempts become instantly visible across the network, making the system robust against fraudulent or unauthorised activities.

Working with blockchain-based storage platforms

Several blockchain-based storage platforms now provide innovative, decentralised and secured backup solutions. For instance, Filecoin allows users to rent out spare storage space in exchange for Filecoin tokens. This incentivises individuals and organisations to contribute their unused storage and create a decentralised storage network. Another example is Storj, which operates by breaking down files into smaller encrypted pieces and distributing them across its network of nodes. Users can rent out their unused storage space and earn cryptocurrency in return, creating a decentralised and efficient storage system.

These platforms utilise blockchain technology to create a trustless and secure environment for data storage, ensuring privacy, integrity, and accessibility.

IPFS (InterPlanetary File System) ***(https://ipfs.io/):*** IPFS is a peer-to-peer hypermedia protocol designed to create a distributed and decentralised method for storing and accessing files. It's not exactly a blockchain but is often used in conjunction with blockchain technology. IPFS is engineered to establish a distributed and decentralised system for file storage and retrieval.

IPFS's core concept is content-addressing, which means files are identified and retrieved based on their unique content, rather than using location-based URLs. At its core, IPFS functions by breaking down files into smaller chunks, and each chunk is assigned a distinct cryptographic hash. This hash acts as a unique identifier for that particular piece of data. Rather than traditional hierarchical file systems, IPFS utilises a Merkle DAG (directed acyclic graph) data structure to link these chunks together, allowing for efficient tracking and retrieval of content.

A key advantage of IPFS lies in its decentralised nature. Instead of relying on a central server, files on IPFS are stored across a network of nodes. These nodes collaborate to store and distribute the chunks of files. When a user requests a specific file, IPFS searches the network for the nodes storing the relevant chunks and retrieves the data, reassembling it on the user's end. One of the critical advantages of IPFS is its resilience against data loss and censorship. Due to the distributed nature of the system, files are replicated across multiple nodes, reducing the risk of losing data if a single node fails. Furthermore, IPFS fosters an environment where information remains available even if certain parts of the network go offline or if there are disruptions.